METRI ENGINEERING SERVICES W.L.L
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TECHNICAL & COMMERCIAL PROPOSAL

for the

Turnkey Project for the Engineering, Procurement, Installation and Commissioning (EPIC) of HVAC Upgrade and Replacement

Submitted to:
QATAR FUEL ADDITIVES COMPANY LIMITED (QAFAC)
Mesaieed Industrial City, Qatar

Submitted by:
METRI ENGINEERING SERVICES (MES)

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1. EXECUTIVE SUMMARY

Metri Engineering Services (MES) is pleased to submit this Technical and Commercial Proposal for the Turnkey Project for the Engineering, Procurement, Installation and Commissioning (EPIC) of HVAC Upgrade and Replacement at QAFAC Methanol/MTBE Facility, Mesaieed Industrial City, Qatar.

This proposal addresses QAFAC's requirement to phase out R-22 refrigerant systems in compliance with the Montreal Protocol and Kigali Amendment, replacing them with environmentally friendly, low-GWP (Global Warming Potential) refrigerant systems meeting QatarEnergy and QCD regulatory requirements.

1.1 Project Highlights

- Full replacement of R-22 HVAC systems across 8 buildings/locations
- A2L low-GWP refrigerant systems (R-454B) meeting ASHRAE 15/34 safety requirements
- 50°C ambient design with 10% additional capacity margin from the existing
- Industrial-grade construction: Stainless steel enclosure, C5 paint grade or equivalent, 7,000-hour salt spray coating
- IE4 Class-H motors (EC)
- Phased execution minimizing operational disruption with temporary cooling solutions
- Comprehensive turnkey delivery including design, supply, installation, commissioning, training, and warranty

1.2 MES Qualifications

MES brings extensive experience in industrial HVAC projects within Qatar's petrochemical sector, including work with QatarEnergy facilities and major industrial clients. Our team is certified for handling A2L refrigerants and hazardous material disposal, with proven capability in executing multi-phase turnkey contracts under strict HSSE requirements.

2. UNDERSTANDING OF THE PROJECT

2.1 Project Background

Qatar Fuel Additives Company (QAFAC) operates a major petrochemical facility in Mesaieed Industrial City producing approximately 3,000 MTPD of Methanol and 1,830 MTPD of MTBE. As part of QAFAC's commitment to environmental responsibility and regulatory compliance, this HVAC upgrade project addresses:

- Montreal Protocol mandate for R-22 refrigerant phase-out
- Requirements for low-GWP refrigerant transition
- QatarEnergy compliance
- Operational excellence through modern, efficient HVAC systems

2.2 Facilities Covered

The project encompasses HVAC replacement across the following 8 buildings/locations:

S/N	Building / Location	Equipment Type	Existing Refrigerant	Status
1	HYPO Building (ME-5201-Shelter)	Packaged AC	R-22	Full Replacement
2	Guard House	Packaged AC	R-22	Full Replacement
3	Substation-4	Packaged AC	R-22	Full Replacement
4	Substation-5	Packaged AC	R-22	Full Replacement
5	Maintenance Workshop #1	ACCU + AHU	R-22	Full Replacement
6	Maintenance Workshop #2	ACCU + AHU	R-22	Full Replacement
7	Firehall Building	ACCU + AHU	R-22	Full Replacement
8	General Services Building	TBC (Site Survey)	R-22	Full Replacement
9	Utility Lab	Packaged AC	R-22	Full Replacement

Note: Existing fresh air treated units shall be retained. Existing control panels, control and power cables, Isolators will be reused where compatible. Temporary AC arrangements shall be provided during replacement phase.

3. TECHNICAL APPROACH

3.1 Design Philosophy

Our design approach ensures full compliance with QAFAC specifications while maximizing system reliability, energy efficiency, and operational flexibility:

3.1.1 Environmental Design Conditions

Parameter	Design Value
Outdoor Design Ambient Temperature	50°C Dry Bulb
Design Capacity Margin	+10% over EXISTING load
Indoor Design Temperature	24°C ± 2°C (as per space requirements)
Relative Humidity	50% ± 5% (for conditioned spaces)

3.1.2 Refrigerant Selection

In compliance with Montreal Protocol, Kigali Amendment, and QCD requirements, we propose the following A2L (mildly flammable) low-GWP refrigerant options:

Refrigerant	GWP	Safety Classification	ODP
R-454B (Proposed)	466	A2L	0
R-22 (Existing - Phased Out)	1,810	A1	0.055

Final refrigerant selection will be confirmed during detailed engineering based on equipment manufacturer recommendations and ASHRAE 15/34 safety requirements for specific building volumes.

3.1.3 Equipment Construction Standards

- **Enclosure:** Stainless Steel (SS 316 grade)
- **Paint System:** C5 grade corrosion protection or equivalent.
- **Coating:** Minimum 7,000 hours salt spray tested
- **Condenser Coils:** Copper tube with Aluminum fins, corrosion resistant treatment
- **Motors: EC** IE4 efficiency class, Class-H insulation, Temperature Rise-B
- **Design Life:** Minimum 15 years
- **Electrical Supply:** 415V - 3Ph - 50Hz

4. PROPOSED EQUIPMENT & SPECIFICATIONS

4.1 Equipment Summary

The following equipment is proposed based on the existing data sheets provided, with 10% increased capacity as specified and transition to eco-friendly refrigerants:

Building	Equipment Type	Proposed Cooling (kW)	Power (kW)	Refrigerant	Qty
Substation-4	PAC	52 (47+10%)	~23	R-454B	1
Substation-5	PAC	43 (39+10%)	~19	R-454B	1
Guard House	PAC	50 (45+10%)	~23	R-454B	1
Utility Lab House	PAC	11 (9.9+10%)	~6	R-454B	1
ME-5201 Shelter	PAC	91 (83+10%)	~45	R-454B	1
Maint. Workshop #1	ACCU	82 (74+10%)	~42	R-454B	1
Maint. Workshop #2	ACCU	104 (94+10%)	~52	R-454B	1
Firehall Building	ACCU	133 (121+10%)	~69	R-454B	1
General Services Bldg	TBC	TBC (Site Survey)	TBC	R-454B	TBC

4.2 Proposed Manufacturer

MES proposes to supply equipment from **PETRA JORDAN**

5. SCOPE OF WORK – INCLUSIONS

MES's turnkey scope includes the following comprehensive deliverables:

5.1 Design Engineering & Planning

- Comprehensive site surveys of all 8 buildings/locations
- Equipment sizing and selection at 50°C ambient with 10% capacity increase
- Detailed HVAC system design incorporating A2L refrigerant safety requirements per ASHRAE 15/34
- Verification and validation of existing electrical infrastructure adequacy (MCC breakers, isolators, control wiring)
- Phase-wise execution plans aligned with QAFAC's long-term HVAC upgrade roadmap
- Temporary cooling design and planning for each building replacement phase
- Preparation of layouts, calculations, and design reports

5.2 Equipment Procurement & Logistics

- Supply of new PACs, AHUs, and ACCUs with low-GWP refrigerants
- Factory testing and inspection
- International freight, customs clearance, and delivery to QAFAC Mesaieed site
- Proper storage and preservation per OEM guidelines

5.3 Temporary Cooling Solutions

- Packaged AC units with ducting and controls during equipment replacement
- Temporary electrical arrangements and power supplies
- Qualified operators to maintain temporary units until permanent systems commissioned

5.4 Dismantling & Safe Disposal

- Safe dismantling of existing R-22 units, refrigerant lines, and electrical connections
- R-22 refrigerant recovery and environmentally compliant disposal per Qatar environmental standards
- Removal of accessories and debris with proper waste management

5.5 Installation & Construction

- Installation of new HVAC units (PACs, AHUs, ACCUs) with duct modifications to the existing duct network.
- New refrigerant piping and insulation
- Electrical power and control cabling with supports
- Cable trays, pull boxes, isolators, and ducting modifications as required
- Replacement of undersized MCCBs, isolators, and wiring modifications where required.

- Utilizing the existing Civil and structural foundations (platforms, supports, foundations)
- New Aluminum cladding for external ducts, pipes, and refrigerant lines
- Restoration works (plastering, cladding, ceiling finishes) where required.

5.6 Testing & Commissioning

- Pressure testing, vacuum testing, leak testing, and refrigerant charging
- Electrical, mechanical, and control function testing
- Performance verification and air balancing
- Commissioning test reports, startup logs.

5.7 Documentation & Handover

- Engineering design calculations and drawings
- As-built drawings
- Warranty certificates
- Operation & Maintenance (O&M) manuals
- Recommended Spare parts list (2 years)

5.8 Training

- 3 days end-user training split between two batches
- Hands-on training for A2L refrigerant handling and safety procedures
- O&M training for new equipment

5.9 Warranty & Support

- 12-month comprehensive warranty post-commissioning for each unit
- On-call support for faults and breakdowns during warranty period

6. SCOPE OF WORK – EXCLUSIONS

The following items are excluded from MES's scope unless specifically agreed otherwise:

6.1 Client Responsibilities (QAFAC)

- Site access permissions, gate passes, and security clearances
- PTW (Permit to Work) processing and approvals
- Coordination with plant operations for building shutdowns
- Power isolation and lockout/tagout procedures
- Safety induction for contractor personnel (provided by QAFAC)
- Review and approval of design documents and MOC submissions

6.2 Infrastructure Exclusions

- Upgrades to main power distribution panels or transformers
- Main building electrical infrastructure modifications beyond local MCC/isolator level
- BMS/SCADA system modifications & integration. By specialist system integrator.
- Fire alarm system modifications & integration. By specialist system integrator.
- Structural reinforcement beyond equipment mounting requirements

6.3 Other Exclusions

- Replacement or modification of existing fresh air treated units (to be retained as specified)
- Existing ductwork replacement (unless found some defective parts during site survey)
- Post-warranty maintenance services (available separately)
- Asbestos or hazardous material abatement (if discovered)
- Costs arising from unforeseen site conditions not visible during site survey

7. EXECUTION METHODOLOGY

7.1 Project Phases

The project will be executed in the following phases to ensure minimal disruption to plant operations:

Phase	Activities	Duration
Phase 1	Mobilization, Site Survey, Detailed Engineering, MOC Submission	6-8 weeks
Phase 2	Equipment Procurement, Manufacturing	12-14 weeks
Phase 3	Delivery, Storage, Building-wise Installation	4-6 weeks
Phase 4	Testing, Commissioning, Training, Handover	2-4 weeks

Note: Outdoor installation works will be paused during summer months (June-September) unless permitted by QAFAC. Project timeline can span multiple execution windows if required.

7.2 Building-Wise Execution Strategy

- One building at a time approach to minimize operational impact
- Temporary cooling deployed before existing unit disconnection
- Pre-fabrication of ducts, piping, supports, and cable trays to minimize on-site work duration
- Weekend/night shift work available for critical areas (subject to PTW approval)

8. PROJECT SCHEDULE

Estimated Overall Project Duration: 24 weeks (excluding summer pause if applicable)

A detailed project schedule will be submitted within 2 weeks of contract award, incorporating:

- QAFAC building shutdown schedule coordination
- Equipment lead times from selected manufacturer
- Summer pause accommodation (June-September)
- Site readiness and PTW availability

9. QUALITY ASSURANCE & HSE

9.1 Quality Management

MES maintains a quality management system. Key QA measures include:

- Inspection and Test Plans (ITP) for all critical activities
- Material and equipment certificates verification
- Site inspections and hold points
- Non-conformance reporting and corrective actions

9.2 Health, Safety & Environment

Full compliance with QAFAC HSSE requirements and PTW system, including:

- Job Hazard Analysis (JHA) for all work activities
- Method statements for critical activities
- Daily toolbox talks and safety briefings
- PPE provision for all personnel
- Medical fitness certificates for all workers
- Certified lifting equipment and personnel
- Environmental compliance for R-22 recovery and disposal
- Noise control and waste management procedures

9.3 Applicable Standards & Codes

Standard/Code	Application
ASHRAE Standards	HVAC design and application
ASHRAE 15/34	A2L refrigerant safety requirements
NFPA 70 & 90A	Electrical safety and ductwork
IEC Standards	Electrical works
QCS	Qatar Construction Specifications
ISO 9001/14001/45001	Contractor QMS, EMS, OH&S certifications
ISO 50001:2018	Energy management requirements
Montreal Protocol	Refrigerant phase-out compliance
Kigali Amendment	Low-GWP refrigerant requirements

10. COMMERCIAL PROPOSAL

10.1 Bill of Quantities

Pricing is provided as lump sum per tender requirements. Detailed breakdown available in attached BOQ spreadsheet.

BILL OF QUANTITIES					
Item	**Description**	**Qty.**	**Unit**	**Rate (QAR)**	**Total Amount (QAR)**
1.0					
1	**Design and Engineering**	Lot	1	90,000	90,000
2	**Supply and Delivery of Materials**				
	1. ME-5201-Shelter		1	198,480	198,480
	2. Guard House		1	133,516	133,516
	3. Substation-4		1	133,516	133,516
	4. Substation-5		1	134,304	134,304
	5. Maintenance Workshop #1		1	210,622	210,622
	6. Maintenance Workshop #2		1	281,340	281,340
	7. Firehall Building		1	299,080	299,080
	8. General Services Building		1	0	0
	9. UTILITY LAB		1	65,358	65,358
3	**Supply and Installation of Temporary Cooling Units**	Lot	1	240,000	240,000
4	**Installation, Testing and Commissioning**		1	0	0
	1. ME-5201-Shelter		1	99,240	99,240
	2. Guard House		1	66,757	66,757
	3. Substation-4		1	66,757	66,757
	4. Substation-5		1	67,152	67,152
	5. Maintenance Workshop #1		1	105,311	105,311
	6. Maintenance Workshop #2		1	140,670	140,670
	7. Firehall Building		1	149,539	149,539
	8. General Services Building		1	0	0
	9. UTILITY LAB		1	32,680	32,680
5	O&M Training and Handling of New Refrigerants	Lot	1	30,000	30,000
6	Dismantling and Safe Disposal of Existing R-22 Units	Lot	1	42,000	42,000
7	Documentation and Handover	Lot	1	12,000	12,000
				TOTAL PRICE Q.R.	**2,598,320**

** General Services Building: There was no any details in the tender document so the price is 0.

10.2 Pricing Notes

1. All prices include applicable taxes, duties, and fees
2. Prices include overhead costs, profit, administrative expenses, utilities, workmen compensation policy, and indirect expenses
3. All direct costs included (materials, labor, machinery)
4. Documentation and reporting costs included (as-built drawings, test reports, JHA, method statements)
5. Delivery, shipping, and transportation to QAFAC Mesaieed Plant included
6. All required PPEs.

10.3 Payment Terms

MES proposes the following milestone-based payment schedule:

Milestone	Description	% of Contract
1	Contract Award / Mobilization Advance	30%
2	Design Approval / Equipment Order	30%
3	Equipment Delivery to Site	15%
4	Installation Complete (per building)	15%
5	Final Commissioning, Training & Handover	5%

11. TERMS & CONDITIONS

11.1 Proposal Validity

This proposal is valid for 90 days from the date of submission.

11.2 Contract Duration

As per QAFAC project planning and site constraints, with accommodation for summer pause periods (June-September) unless specifically permitted.

11.3 Warranty Period

- 12-month comprehensive warranty for each unit, commencing from successful installation and commissioning (not delivery)
- On-call support for faults and breakdowns included during warranty period

11.4 Force Majeure

Standard force majeure provisions apply for events beyond reasonable control of either party.

11.5 Compliance

MES confirms full compliance with all QAFAC HSSE requirements, PTW system, and applicable Qatar regulations. All installations will comply with Recognized and Generally Accepted Good Engineering Practices.

Best Regards,

METRI ENGINEERING SERVICES W.L.L.

Eng. Fayez Metri

PETRA AIR TO AIR SELECTION

Version 4.2

Version Date : 07-01-2025

17/12/2025

Project name : QAFAC different location

Study Engineer : Mohammad Awad

Model Ref. : PAC-04 Sub-station R1 (PPHb-195) **Qty :** 1

Input Data :

Power Supply		:	380-415/3ph/50Hz			
Required capacity	**[kW]**	:	51.70	**[Btu/hr]**	:	176400.40
Ambient	**[°C]**	:	50.00	**[°F]**	:	122.00
Supply Fan Air Flow Rate	**[L/s]**	:	2800	**[CFM]**	:	5933
Ent. dbt	**[°C]**	:	29.20	**[°F]**	:	84.56
Ent. wbt	**[°C]**	:	21.00	**[°F]**	:	69.80

Output Data :

Leaving dbt	**[°C]**	:	16.82	**[°F]**	:	62.28
Leaving wbt	**[°C]**	:	15.73	**[°F]**	:	60.31
APD dry	**[Pa]**	:	64	**[inwg]**	:	0.257
APD wet	**[Pa]**	:	83	**[inwg]**	:	0.333
Calculated capacity	**[kW]**	:	55.60	**[Btu/hr]**	:	189707
Sensible capacity	**[kW]**	:	42.80	**[Btu/hr]**	:	146034
Face velocity	**[m/s]**	:	2.36	**[fpm]**	:	465
No. of tubes		:	34			
Compressor Power Input	**[kW]**	:	17.71			

Refrigeration Data :

Refrigerant : R454B

Evaporator Data :

>> Coil :

Area	**[m²]**	:	1.18	**[ft²]**	:	12.70
Fins Per Meter		:	472	**Fins Per Inch**	:	12
Rows		:	4			

Condenser Data :

>> Coil :

No. of condensers		:	1			
Fins Per Meter		:	472	**Fins Per Inch**	:	12
Condenser rows		:	4			
Total area	**[m²]**	:	2.89	**[ft²]**	:	31.14

>> Fan :

No. of fans : 3

Compressor :

No. : 2

PETRA AIR TO AIR SELECTION

Version 4.2

Version Date : 07-01-2025

17/12/2025

Project name : QAFAC different location

Study Engineer : Mohammad Awad

Model Ref. : PAC-05 Sub-station R1 (PPHb-230) **Qty :** 1

Input Data :

Power Supply		:	380-415/3ph/50Hz			
Required capacity	**[kW]**	:	51.70	**[Btu/hr]**	:	176400.40
Ambient	**[°C]**	:	50.00	**[°F]**	:	122.00
Supply Fan Air Flow Rate	**[L/s]**	:	1800	**[CFM]**	:	3814
Ent. dbt	**[°C]**	:	30.20	**[°F]**	:	86.36
Ent. wbt	**[°C]**	:	22.20	**[°F]**	:	71.96

Output Data :

Leaving dbt	**[°C]**	:	15.39	**[°F]**	:	59.70
Leaving wbt	**[°C]**	:	14.37	**[°F]**	:	57.87
APD dry	**[Pa]**	:	55	**[inwg]**	:	0.219
APD wet	**[Pa]**	:	71	**[inwg]**	:	0.285
Calculated capacity	**[kW]**	:	53.40	**[Btu/hr]**	:	182201
Sensible capacity	**[kW]**	:	32.90	**[Btu/hr]**	:	112255
Face velocity	**[m/s]**	:	2.15	**[fpm]**	:	423
No. of tubes		:	24			
Compressor Power Input	**[kW]**	:	20.13			

Refrigeration Data :

Refrigerant : R454B

Evaporator Data :

>> Coil :

Area	**[m²]**	:	0.84	**[ft²]**	:	9.04
Fins Per Meter		:	472	**Fins Per Inch**	:	12
Rows		:	4			

Condenser Data :

>> Coil :

No. of condensers		:	1			
Fins Per Meter		:	551	**Fins Per Inch**	:	14
Condenser rows		:	4			
Total area	**[m²]**	:	2.89	**[ft²]**	:	31.14

>> Fan :

No. of fans : 3

Compressor :

No. : 2

PETRA AIR TO AIR SELECTION

Version 4.2

Version Date : 07-01-2025

17/12/2025

Project name : QAFAC different location

Study Engineer : Mohammad Awad

Model Ref. : PAC-GH-001 R1. (PPHb-195) **Qty :** 1

Input Data :

Power Supply		:	380-415/3ph/50Hz			
Required capacity	**[kW]**	:	49.50	**[Btu/hr]**	:	168894.00
Ambient	**[°C]**	:	50.00	**[°F]**	:	122.00
Supply Fan Air Flow Rate	**[L/s]**	:	2200	**[CFM]**	:	4662
Ent. dbt	**[°C]**	:	28.50	**[°F]**	:	83.30
Ent. wbt	**[°C]**	:	21.90	**[°F]**	:	71.42

Output Data :

Leaving dbt	**[°C]**	:	16.23	**[°F]**	:	61.21
Leaving wbt	**[°C]**	:	15.43	**[°F]**	:	59.77
APD dry	**[Pa]**	:	43	**[inwg]**	:	0.171
APD wet	**[Pa]**	:	55	**[inwg]**	:	0.223
Calculated capacity	**[kW]**	:	54.70	**[Btu/hr]**	:	186636
Sensible capacity	**[kW]**	:	33.30	**[Btu/hr]**	:	113620
Face velocity	**[m/s]**	:	1.86	**[fpm]**	:	366
No. of tubes		:	34			
Compressor Power Input	**[kW]**	:	17.51			

Refrigeration Data :

Refrigerant : R454B

Evaporator Data :

>> Coil :

Area	**[m²]**	:	1.18	**[ft²]**	:	12.70
Fins Per Meter		:	472	**Fins Per Inch**	:	12
Rows		:	4			

Condenser Data :

>> Coil :

No. of condensers		:	1			
Fins Per Meter		:	551	**Fins Per Inch**	:	14
Condenser rows		:	4			
Total area	**[m²]**	:	2.89	**[ft²]**	:	31.14

>> Fan :

No. of fans : 3

Compressor :

No. : 2

PETRA AIR TO AIR SELECTION

Version 4.2

Version Date : 07-01-2025

17/12/2025

Project name : QAFAC different location

Study Engineer : Mohammad Awad

Model Ref. : PAC-UT-001 R1 (PPHb-85) **Qty :** 1

Input Data :

Power Supply		:	380-415/3ph/50Hz			
Required capacity	**[kW]**	:	10.90	**[Btu/hr]**	:	37190.80
Ambient	**[°C]**	:	50.00	**[°F]**	:	122.00
Supply Fan Air Flow Rate	**[L/s]**	:	450	**[CFM]**	:	954
Ent. dbt	**[°C]**	:	30.00	**[°F]**	:	86.00
Ent. wbt	**[°C]**	:	20.50	**[°F]**	:	68.90

Output Data :

Leaving dbt	**[°C]**	:	14.58	**[°F]**	:	58.24
Leaving wbt	**[°C]**	:	13.36	**[°F]**	:	56.05
APD dry	**[Pa]**	:	55	**[inwg]**	:	0.219
APD wet	**[Pa]**	:	71	**[inwg]**	:	0.285
Calculated capacity	**[kW]**	:	11.50	**[Btu/hr]**	:	39238
Sensible capacity	**[kW]**	:	8.60	**[Btu/hr]**	:	29343
Face velocity	**[m/s]**	:	2.15	**[fpm]**	:	423
No. of tubes		:	18			
Compressor Power Input	**[kW]**	:	4.11			

Refrigeration Data :

Refrigerant : R454B

Evaporator Data :

>> Coil :

Area	**[m²]**	:	0.21	**[ft²]**	:	2.26
Fins Per Meter		:	472	**Fins Per Inch**	:	12
Rows		:	4			

Condenser Data :

>> Coil :

No. of condensers		:	1			
Fins Per Meter		:	472	**Fins Per Inch**	:	12
Condenser rows		:	3			
Total area	**[m²]**	:	1.25	**[ft²]**	:	13.50

>> Fan :

No. of fans : 2

Compressor :

No. : 1

PETRA AIR TO AIR SELECTION

Version 4.2

Version Date : 07-01-2025

17/12/2025

Project name : QAFAC different location

Study Engineer : Mohammad Awad

Model Ref. : PAC-59-001 R1. (PPHb-360) **Qty :** 1

Input Data :

Power Supply		:	380-415/3ph/50Hz			
Required capacity	**[kW]**	:	91.30	**[Btu/hr]**	:	311515.60
Ambient	**[°C]**	:	50.00	**[°F]**	:	122.00
Supply Fan Air Flow Rate	**[L/s]**	:	4147	**[CFM]**	:	8788
Ent. dbt	**[°C]**	:	27.50	**[°F]**	:	81.50
Ent. wbt	**[°C]**	:	19.00	**[°F]**	:	66.20

Output Data :

Leaving dbt	**[°C]**	:	12.62	**[°F]**	:	54.72
Leaving wbt	**[°C]**	:	12.44	**[°F]**	:	54.39
APD dry	**[Pa]**	:	84	**[inwg]**	:	0.337
APD wet	**[Pa]**	:	109	**[inwg]**	:	0.438
Calculated capacity	**[kW]**	:	92.60	**[Btu/hr]**	:	315951
Sensible capacity	**[kW]**	:	76.10	**[Btu/hr]**	:	259653
Face velocity	**[m/s]**	:	1.76	**[fpm]**	:	346
No. of tubes		:	48			
Compressor Power Input	**[kW]**	:	29.25			

Refrigeration Data :

Refrigerant : R454B

Evaporator Data :

>> Coil :

Area	**[m²]**	:	2.35	**[ft²]**	:	25.30
Fins Per Meter		:	551	**Fins Per Inch**	:	14
Rows		:	8			

Condenser Data :

>> Coil :

No. of condensers		:	1			
Fins Per Meter		:	551	**Fins Per Inch**	:	14
Condenser rows		:	4			
Total area	**[m²]**	:	3.93	**[ft²]**	:	42.25

>> Fan :

No. of fans : 2

Compressor :

No. : 2

PETRA AIR TO AIR SELECTION

Version 4.2

Version Date : 07-01-2025

17/12/2025

Project name : QAFAC different location

Study Engineer : Mohammad Awad

Model Ref. : ACCU-01M R1 (DSPb-360+PAHHC 80) **Qty :** 1

Input Data :

Power Supply		:	380-415/3ph/50Hz			
Required capacity	**[kW]**	:	81.62	**[Btu/hr]**	:	278487.40
Ambient	**[°C]**	:	50.00	**[°F]**	:	122.00
Supply Fan Air Flow Rate	**[L/s]**	:	3500	**[CFM]**	:	7417
Ent. dbt	**[°C]**	:	30.00	**[°F]**	:	86.00
Ent. wbt	**[°C]**	:	22.00	**[°F]**	:	71.60

Output Data :

Leaving dbt	**[°C]**	:	16.70	**[°F]**	:	62.06
Leaving wbt	**[°C]**	:	15.65	**[°F]**	:	60.17
APD dry	**[Pa]**	:	60	**[inwg]**	:	0.242
APD wet	**[Pa]**	:	78	**[inwg]**	:	0.315
Calculated capacity	**[kW]**	:	85.70	**[Btu/hr]**	:	292408
Sensible capacity	**[kW]**	:	57.50	**[Btu/hr]**	:	196190
Face velocity	**[m/s]**	:	2.28	**[fpm]**	:	449
No. of tubes		:	44			
Compressor Power Input	**[kW]**	:	28.84			

Refrigeration Data :

Refrigerant : R454B

Evaporator Data :

>> Coil :

Area	**[m²]**	:	1.53	**[ft²]**	:	16.47
Fins Per Meter		:	472	**Fins Per Inch**	:	12
Rows		:	4			

Condenser Data :

>> Coil :

No. of condensers		:	1			
Fins Per Meter		:	551	**Fins Per Inch**	:	14
Condenser rows		:	4			
Total area	**[m²]**	:	3.93	**[ft²]**	:	42.25

>> Fan :

No. of fans : 2

Compressor :

No. : 2

PETRA AIR TO AIR SELECTION

Version 4.2

Version Date : 07-01-2025

17/12/2025

Project name : QAFAC different location

Study Engineer : Mohammad Awad

Model Ref. : ACCU-02M R1 (DSPb-450+PAHHC 100) **Qty :** 1

Input Data :

Power Supply		:	380-415/3ph/50Hz			
Required capacity	**[kW]**	:	103.51	**[Btu/hr]**	:	353176.10
Ambient	**[°C]**	:	50.00	**[°F]**	:	122.00
Supply Fan Air Flow Rate	**[L/s]**	:	4500	**[CFM]**	:	9536
Ent. dbt	**[°C]**	:	30.00	**[°F]**	:	86.00
Ent. wbt	**[°C]**	:	22.00	**[°F]**	:	71.60

Output Data :

Leaving dbt	**[°C]**	:	16.98	**[°F]**	:	62.56
Leaving wbt	**[°C]**	:	15.95	**[°F]**	:	60.71
APD dry	**[Pa]**	:	55	**[inwg]**	:	0.222
APD wet	**[Pa]**	:	72	**[inwg]**	:	0.288
Calculated capacity	**[kW]**	:	105.50	**[Btu/hr]**	:	359966
Sensible capacity	**[kW]**	:	72.40	**[Btu/hr]**	:	247029
Face velocity	**[m/s]**	:	2.17	**[fpm]**	:	427
No. of tubes		:	46			
Compressor Power Input	**[kW]**	:	33.58			

Refrigeration Data :

Refrigerant : R454B

Evaporator Data :

>> Coil :

Area	**[m²]**	:	2.08	**[ft²]**	:	22.39
Fins Per Meter		:	472	**Fins Per Inch**	:	12
Rows		:	4			

Condenser Data :

>> Coil :

No. of condensers		:	2			
Fins Per Meter		:	551	**Fins Per Inch**	:	14
Condenser rows		:	4			
Total area	**[m²]**	:	5.75	**[ft²]**	:	61.84

>> Fan :

No. of fans : 2

Compressor :

No. : 2

PETRA AIR TO AIR SELECTION

Version 4.2

Version Date : 07-01-2025

17/12/2025

Project name : QAFAC different location

Study Engineer : Mohammad Awad

Model Ref. : ACCU-FR R1 (DSPb-530+PAHHC 100) **Qty :** 1

Input Data :

Power Supply		:	380-415/3ph/50Hz			
Required capacity	**[kW]**	:	133.00	**[Btu/hr]**	:	453796.00
Ambient	**[°C]**	:	50.00	**[°F]**	:	122.00
Supply Fan Air Flow Rate	**[L/s]**	:	4500	**[CFM]**	:	9536
Ent. dbt	**[°C]**	:	30.00	**[°F]**	:	86.00
Ent. wbt	**[°C]**	:	22.00	**[°F]**	:	71.60

Output Data :

Leaving dbt	**[°C]**	:	14.60	**[°F]**	:	58.28
Leaving wbt	**[°C]**	:	14.08	**[°F]**	:	57.34
APD dry	**[Pa]**	:	68	**[inwg]**	:	0.273
APD wet	**[Pa]**	:	89	**[inwg]**	:	0.356
Calculated capacity	**[kW]**	:	133.70	**[Btu/hr]**	:	456184
Sensible capacity	**[kW]**	:	85.60	**[Btu/hr]**	:	292067
Face velocity	**[m/s]**	:	2.17	**[fpm]**	:	427
No. of tubes		:	46			
Compressor Power Input	**[kW]**	:	48.34			

Refrigeration Data :

Refrigerant : R454B

Evaporator Data :

>> Coil :

Area	**[m²]**	:	2.08	**[ft²]**	:	22.39
Fins Per Meter		:	394	**Fins Per Inch**	:	10
Rows		:	6			

Condenser Data :

>> Coil :

No. of condensers		:	2			
Fins Per Meter		:	551	**Fins Per Inch**	:	14
Condenser rows		:	4			
Total area	**[m²]**	:	6.37	**[ft²]**	:	68.61

>> Fan :

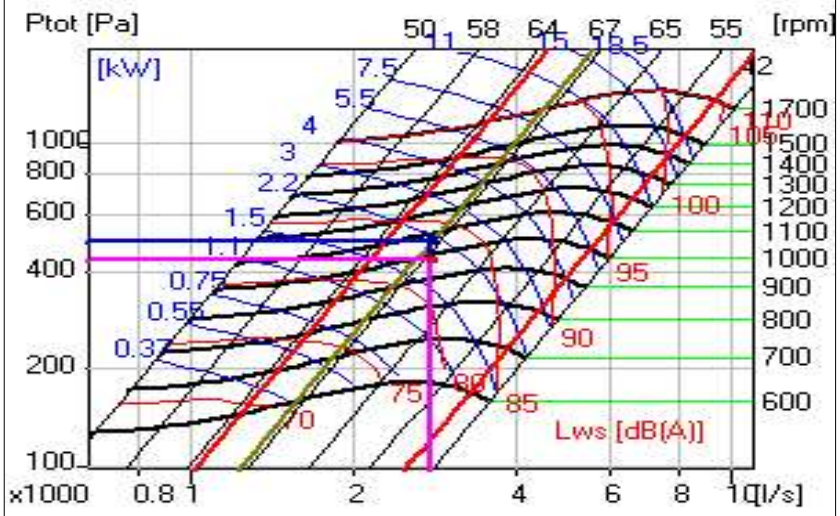
No. of fans : 2

Compressor :

No. : 2

		Project Reference	EPIC for HVAC upgrade & replac PAC-04 Substation

12-12 Forward
Double Unit - Ducted Outlet



Working Point Data		
Total EFFICIENCY	66.6	%
Static Efficiency	58	%
Static Pressure	435	Pa
Velocity Pressure	65	Pa
Total Pressure	500	Pa
Fan Power	2.101	kW
Motor Power	2.731	kW
Volume Flow	2800	l/s
Air Velocity	10.4	m/s
Fan Speed	1063	rpm
Temperature	69.	°F
Altitude	0.	ft
Density	1.2	kg/m³
Motor Efficiency	85	%
Specific Fan Power	0.883	kW/(m³/s)

Operational Limits		
Max. Power	5.5	kW
Max. Fan Speed	1400	rpm
Min. Temperature	-5	°F
Max. Temperature	185	°F

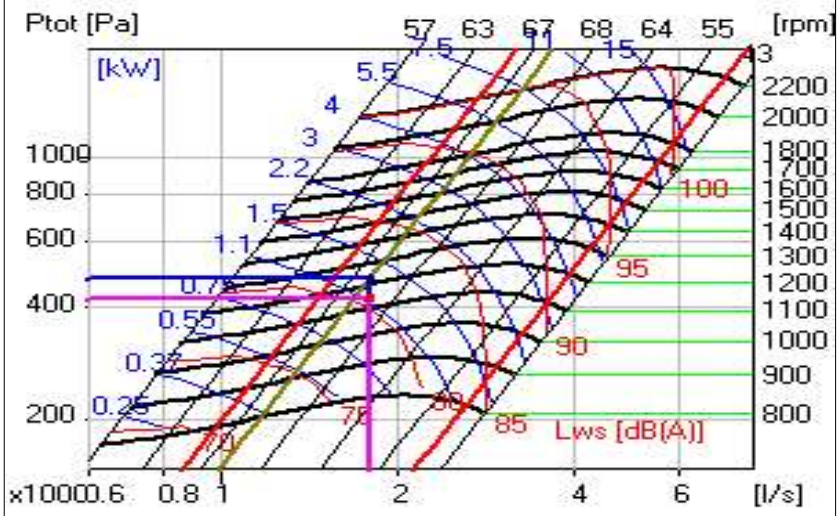
	Sound Power Level [dB]								Lw-tot		LwA-tot	
Hz	63	125	250	500	1000	2000	4000	8000				
Inlet side External - Lwi	79.6	85.1	82	76.6	78	77.6	74.7	68	88.9	dB	83.7	dBA
Outlet side in-duct - Lwo	85.9	89.2	84.8	79.3	79	78.9	76	69.2	92.6	dB	85.4	dBA
Outlet side Ext. - Lwom	73.9	82.1	81.6	78.2	78.7	78.8	76	69.2	87.8	dB	84.5	dBA

Notes

2 FAN - 1 MOTOR
MOTOR SIZE = 3.0 KW
ESP= 200 Pa

		Project Reference	EPIC for HVAC upgrade & replac PAC-05 Substation

10-10 Forward
Double Unit - Ducted Outlet



Working Point Data

Total EFFICIENCY	67.1	%
Static Efficiency	59.6	%
Static Pressure	422	Pa
Velocity Pressure	53	Pa
Total Pressure	475	Pa
Fan Power	1.274	kW
Motor Power	1.657	kW
Volume Flow	1800	l/s
Air Velocity	9.4	m/s
Fan Speed	1226	rpm
Temperature	69.	°F
Altitude	0.	ft
Density	1.2	kg/m³
Motor Efficiency	85	%
Specific Fan Power	0.833	kW/(m³/s)

Operational Limits

Max. Power	3	kW
Max. Fan Speed	1800	rpm
Min. Temperature	-5	°F
Max. Temperature	185	°F

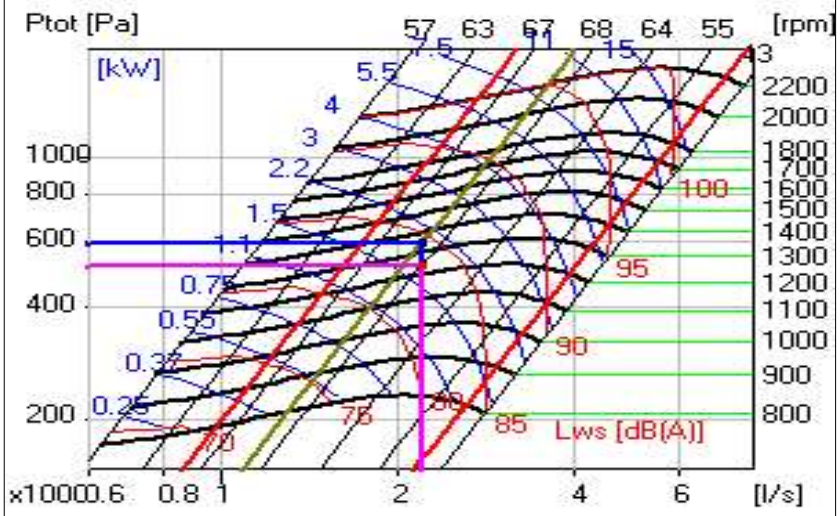
	Sound Power Level [dB]								Lw-tot		LwA-tot	
Hz	63	125	250	500	1000	2000	4000	8000				
Inlet side External - Lwi	77.8	79.3	80.6	74.7	76.3	75.4	72.1	66	85.9	dB	81.6	dBA
Outlet side in-duct - Lwo	85.6	84.5	84.2	77.7	77.6	76.8	73.7	67.5	90.4	dB	83.6	dBA
Outlet side Ext. - Lwom	72.4	76.3	80.2	76.2	77.1	76.7	73.7	67.5	85.2	dB	82.6	dBA

Notes

2 FAN - 1 MOTOR
MOTOR SIZE = 2.2 KW
ESP= 200 Pa

		Project Reference	EPIC for HVAC upgrade & replac PAC-GH-001

10-10 Forward
Double Unit - Ducted Outlet



Working Point Data

Total EFFICIENCY	67.6	%
Static Efficiency	58.6	%
Static Pressure	514	Pa
Velocity Pressure	79	Pa
Total Pressure	593	Pa
Fan Power	1.931	kW
Motor Power	2.51	kW
Volume Flow	2200	l/s
Air Velocity	11.5	m/s
Fan Speed	1346	rpm
Temperature	69.	°F
Altitude	0.	ft
Density	1.2	kg/m³
Motor Efficiency	85	%
Specific Fan Power	1.032	kW/(m³/s)

Operational Limits

Max. Power	3	kW
Max. Fan Speed	1800	rpm
Min. Temperature	-5	°F
Max. Temperature	185	°F

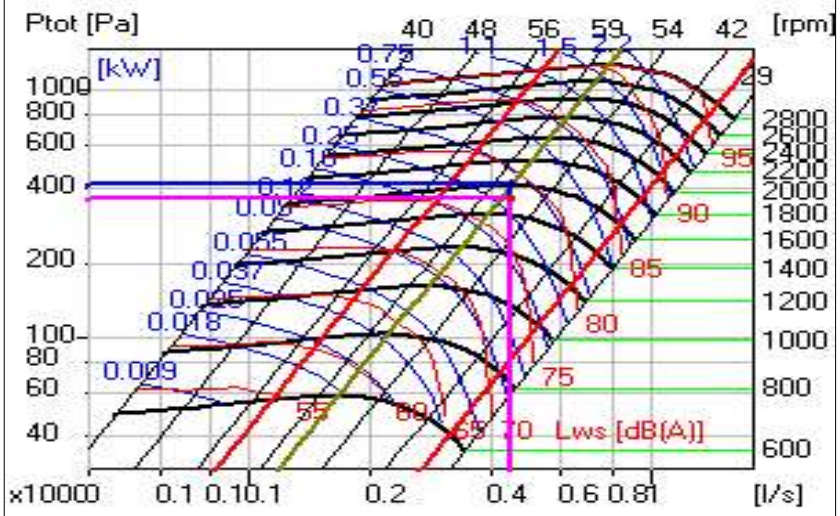
	Sound Power Level [dB]								Lw-tot		LwA-tot	
Hz	63	125	250	500	1000	2000	4000	8000				
Inlet side External - Lwi	80.1	79.6	83.1	77.2	78.7	78.5	76	70.9	88.1	dB	84.5	dBA
Outlet side in-duct - Lwo	88	84.9	86.6	80.1	80.3	79.8	77.4	72.2	92.5	dB	86.4	dBA
Outlet side Ext. - Lwom	74.7	76.7	82.6	78.6	79.8	79.6	77.4	72.2	87.7	dB	85.5	dBA

Notes

2 FAN - 1 MOTOR
MOTOR SIZE = 3.0 KW
ESP= 300 Pa

		Project Reference	EPIC for HVAC upgrade & replac PAC-UT-001

7-7 Forward
Single Unit - Ducted Outlet



Working Point Data

Total EFFICIENCY	56.8	%
Static Efficiency	49.7	%
Static Pressure	366	Pa
Velocity Pressure	52	Pa
Total Pressure	418	Pa
Fan Power	0.332	kW
Motor Power	0.431	kW
Volume Flow	450	l/s
Air Velocity	9.3	m/s
Fan Speed	1614	rpm
Temperature	69.	°F
Altitude	0.	ft
Density	1.2	kg/m³
Motor Efficiency	85	%
Specific Fan Power	0.867	kW/(m³/s)

Operational Limits

Max. Power	3	kW
Max. Fan Speed	2600	rpm
Min. Temperature	-5	°F
Max. Temperature	185	°F

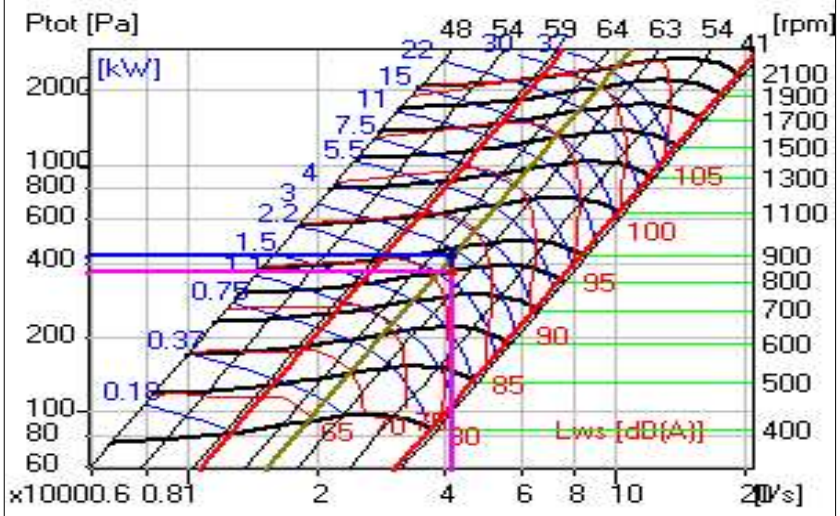
	Sound Power Level [dB]								Lw-tot		LwA-tot	
Hz	63	125	250	500	1000	2000	4000	8000				
Inlet side External - Lwi	71.7	71.7	79.5	68.1	71.8	73.8	70.8	67.1	82.6	dB	79.1	dBA
Outlet side in-duct - Lwo	80.6	77.9	83.5	71.1	73.8	74.8	72.1	68.3	86.9	dB	81.1	dBA
Outlet side Ext. - Lwom	64.6	67.2	77.6	68.6	73	74.6	72	68.3	81.6	dB	79.7	dBA

Notes

MOTOR SIZE = 0.55 KW
ESP= 150 Pa

		Project Reference	EPIC for HVAC upgrade & replac PAC-59-001

355 Forward
Double Unit - Ducted Outlet



Working Point Data		
Total EFFICIENCY	63.8	%
Static Efficiency	54.7	%
Static Pressure	367	Pa
Velocity Pressure	61	Pa
Total Pressure	428	Pa
Fan Power	2.782	kW
Motor Power	3.617	kW
Volume Flow	4147	l/s
Air Velocity	10.1	m/s
Fan Speed	865	rpm
Temperature	69.	°F
Altitude	0.	ft
Density	1.2	kg/m³
Motor Efficiency	85	%
Specific Fan Power	0.789	kW/(m³/s)

Operational Limits		
Max. Power	15	kW
Max. Fan Speed	1700	rpm
Min. Temperature	-5	°F
Max. Temperature	212	°F

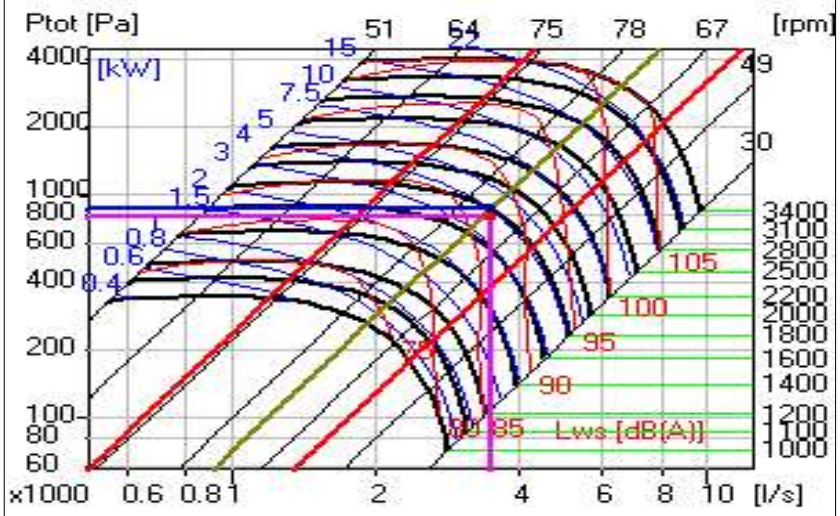
	Sound Power Level [dB]								Lw-tot		LwA-tot	
Hz	63	125	250	500	1000	2000	4000	8000				
Inlet side External - Lwi	80.5	85.2	78.5	78.3	76.3	75.5	72.4	68.2	88.4	dB	82.3	dBA
Outlet side in-duct - Lwo	89.8	90.1	80.4	78.9	76.5	75.6	72.4	68.2	93.6	dB	83.1	dBA
Outlet side Ext. - Lwom	79.5	84.4	78	78.1	76.2	75.5	72.4	68.2	87.7	dB	82.1	dBA

Notes

2 FAN - 1 MOTOR
MOTOR SIZE = 4.0 KW
ESP= 450 Pa

		Project Reference	EPIC for HVAC upgrade & replac ACCU-01M

450 Backward
Single Unit - Ducted Outlet



Working Point Data

Total EFFICIENCY	74.1	%
Static Efficiency	68	%
Static Pressure	784	Pa
Velocity Pressure	70	Pa
Total Pressure	854	Pa
Fan Power	4.035	kW
Motor Power	5.245	kW
Volume Flow	3500	l/s
Air Velocity	10.8	m/s
Fan Speed	1805	rpm
Temperature	69.	°F
Altitude	0.	ft
Density	1.2	kg/m³
Motor Efficiency	85	%
Specific Fan Power	1.356	kW/(m³/s)
FEG	90	

Operational Limits

Max. Power	15	kW
Max. Fan Speed	2800	rpm
Min. Temperature	-5	°F
Max. Temperature	212	°F

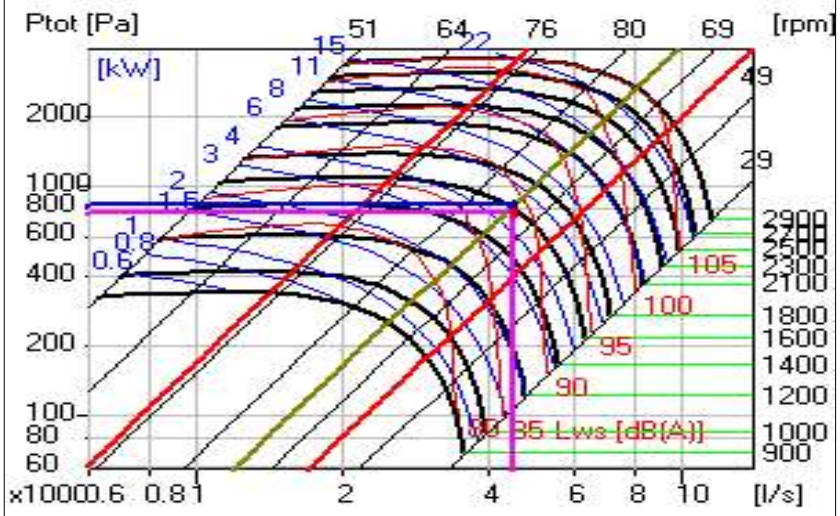
	Sound Power Level [dB]								Lw-tot		LwA-tot	
Hz	63	125	250	500	1000	2000	4000	8000				
Inlet side External - Lwi	83.6	83.8	87.3	85.8	81.7	78.8	73.7	66.5	92.2	dB	87.4	dBA
Outlet side in-duct - Lwo	94.5	89.5	89.4	87.8	87.4	82.7	75.8	67.7	97.8	dB	91.1	dBA
Outlet side Ext. - Lwom	85.8	85.1	87.7	87.3	87.2	82.6	75.8	67.7	94.1	dB	90.7	dBA

Notes

MOTOR SIZE = 5.5 KW
ESP= 450 Pa

		Project Reference	EPIC for HVAC upgrade & replac ACCU-02M

500 Backward
Single Unit - Ducted Outlet



Working Point Data		
Total EFFICIENCY	75	%
Static Efficiency	68.3	%
Static Pressure	757	Pa
Velocity Pressure	73	Pa
Total Pressure	830	Pa
Fan Power	4.985	kW
Motor Power	6.481	kW
Volume Flow	4500	l/s
Air Velocity	11.1	m/s
Fan Speed	1624	rpm
Temperature	69.	°F
Altitude	0.	ft
Density	1.2	kg/m³
Motor Efficiency	85	%
Specific Fan Power	1.303	kW/(m³/s)
FEG	85	

Operational Limits		
Max. Power	15	kW
Max. Fan Speed	2350	rpm
Min. Temperature	-5	°F
Max. Temperature	212	°F

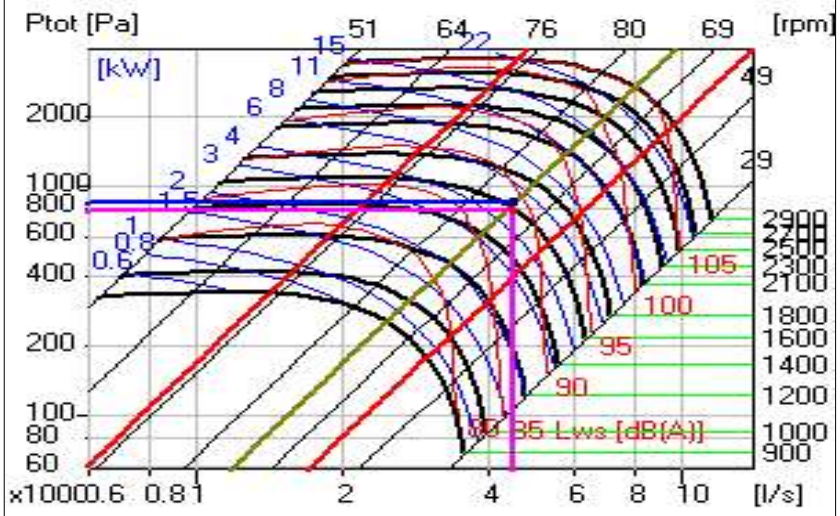
	Sound Power Level [dB]								Lw-tot		LwA-tot	
Hz	63	125	250	500	1000	2000	4000	8000				
Inlet side External - Lwi	84.3	86.5	92.1	86.5	82.2	79.8	74.4	69.5	94.9	dB	89	dBA
Outlet side in-duct - Lwo	94.3	92.5	94.7	89.3	88	83.3	76.5	70.6	99.6	dB	92.7	dBA
Outlet side Ext. - Lwom	86.4	88.7	93.3	88.9	87.8	83.2	76.4	70.6	96.9	dB	92.2	dBA

Notes

MOTOR SIZE = 7.5 KW
ESP= 450 Pa

		Project Reference	EPIC for HVAC upgrade & replac ACCU-FR

500 Backward
Single Unit - Ducted Outlet



Working Point Data		
Total EFFICIENCY	75.3	%
Static Efficiency	68.8	%
Static Pressure	776	Pa
Velocity Pressure	73	Pa
Total Pressure	849	Pa
Fan Power	5.078	kW
Motor Power	6.601	kW
Volume Flow	4500	l/s
Air Velocity	11.1	m/s
Fan Speed	1636	rpm
Temperature	69.	°F
Altitude	0.	ft
Density	1.2	kg/m³
Motor Efficiency	85	%
Specific Fan Power	1.328	kW/(m³/s)
FEG	85	

Operational Limits		
Max. Power	15	kW
Max. Fan Speed	2350	rpm
Min. Temperature	-5	°F
Max. Temperature	212	°F

	Sound Power Level [dB]								Lw-tot		LwA-tot	
Hz	63	125	250	500	1000	2000	4000	8000				
Inlet side External - Lwi	84.3	86.7	92.1	86.7	82.3	79.9	74.5	69.7	95	dB	89.1	dBA
Outlet side in-duct - Lwo	94.5	92.7	94.6	89.5	88	83.4	76.6	70.8	99.7	dB	92.8	dBA
Outlet side Ext. - Lwom	86.5	88.8	93.2	89	87.9	83.3	76.6	70.8	97	dB	92.2	dBA

Notes

MOTOR SIZE = 7.5 KW
ESP= 450 Pa